

SOLIDWORKS API (.NET VBA C#) — Reference (192/337): IModelDoc2 (part 15 of 17)

SOLIDWORKS API Documentation (converted from CHM)

IModelDoc2 (part 15 of 17) (Type)

SOLIDWORKS API Help

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SetUserPreferenceStringValue Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SetUserPreferenceStringValue Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::SetUserPreferenceString](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetUserPreferenceStringValue( _  
    ByVal UserPreference As System.Integer, _  
    ByVal Value As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UserPreference As System.Integer  
Dim Value As System.String  
Dim value As System.Boolean  
  
value = instance.SetUserPreferenceStringValue(UserPreference, Value)
```

C#

```
System.bool SetUserPreferenceStringValue(  
    System.int UserPreference,  
    System.string Value  
)
```

C++/CLI

```
System.bool SetUserPreferenceStringValue(  
& System.int UserPreference,  
& System.String^ Value  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPreference

User preference value as defined in [swUserPreferenceStringValue_e](#)

Value

Value of the user preference specified in UserPreference

Return Value

True if user preference value is changed, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SetUserPreferenceStringValue](#).

Remarks

This method is equivalent to interactively setting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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SetUserPreferenceTextFormat Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SetUserPreferenceTextFormat Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::SetUserPreferenceTextFormat](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetUserPreferenceTextFormat( _  
    ByVal UserPreferenceValue As System.Integer, _  
    ByVal Value As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UserPreferenceValue As System.Integer  
Dim Value As System.Object  
Dim value As System.Boolean  
  
value = instance.SetUserPreferenceTextFormat(UserPreferenceValue, Value)
```

C#

```
System.bool SetUserPreferenceTextFormat(  
    System.int UserPreferenceValue,  
    System.object Value  
)
```

C++/CLI

```
System.bool SetUserPreferenceTextFormat(  
& System.int UserPreferenceValue,  
& System.Object^ Value  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPreferenceValue

User preference to change as defined in [swUserPreferenceTextFormat_e](#)

Value

[Text format](#) to which to assign the user preference specified in UserPreferenceValue

Return Value

True if the setting is changed successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SetUserPreferenceTextFormat](#).

Remarks

This method is equivalent to interactively setting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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SetUserPreferenceToggle Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SetUserPreferenceToggle Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::SetUserPreferenceToggle](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetUserPreferenceToggle( _  
    ByVal UserPreferenceValue As System.Integer, _  
    ByVal OnFlag As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UserPreferenceValue As System.Integer  
Dim OnFlag As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SetUserPreferenceToggle(UserPreferenceValue, OnFlag)
```

C#

```
System.bool SetUserPreferenceToggle(  
    System.int UserPreferenceValue,  
    System.bool OnFlag  
)
```

C++/CLI

```
System.bool SetUserPreferenceToggle(  
& System.int UserPreferenceValue,  
& System.bool OnFlag  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPreferenceValue

Value to toggle as defined in [swUserPreferenceToggle_e](#)

OnFlag

True to toggle the value on, false to toggle the value off

Return Value

True if the toggle is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SetUserPreferenceToggle](#).

Example

[Hide or Show All Types \(VBA\)](#)
[Ignore Feature Colors \(VBA\)](#)

Remarks

This method is equivalent to interactively setting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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SetZebraStripeData Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SetZebraStripeData Method (IModelDoc2)

Sets the zebra-line data.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetZebraStripeData(  
    ByVal Size As System.Double, _  
    ByVal Ratio As System.Double, _  
    ByVal Color1 As System.Integer, _  
    ByVal Color2 As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Size As System.Double  
Dim Ratio As System.Double  
Dim Color1 As System.Integer  
Dim Color2 As System.Integer  
  
instance.SetZebraStripeData(Size, Ratio, Color1, Color2)
```

C#

```
void SetZebraStripeData(  
    System.double Size,  
    System.double Ratio,  
    System.int Color1,  
    System.int Color2  
)
```

C++/CLI

```
void SetZebraStripeData(  
& System.double Size,  
& System.double Ratio,  
& System.int Color1,  
& System.int Color2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Size
Number of stripes

Ratio
Width of the stripes

Color1
First color in zebra stripe design

Color2
Second color in zebra stripe design

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SetZebraStripeData](#).

Remarks

The Size parameter is inversely related; a large size value generates lots of small stripes.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetZebraStripeData Method](#)
[IModelView::DisplayZebraStripes Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ShowComponent2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ShowComponent2 Method (IModelDoc2)

Shows the selected component.

.NET Syntax

Visual Basic (Declaration)

```
Sub ShowComponent2()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ShowComponent2()
```

C#

```
void ShowComponent2()
```

C++/CLI

```
void ShowComponent2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ShowComponent2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::HideComponent2 Method](#)
[IModelDoc2::IsActive Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ShowConfiguration Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ShowConfiguration Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ShowConfiguration2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ShowConfiguration( _  
    ByVal ConfigurationName As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ConfigurationName As System.String  
  
instance.ShowConfiguration(ConfigurationName)
```

C#

```
void ShowConfiguration(  
    System.string ConfigurationName  
)
```

C++/CLI

```
void ShowConfiguration(  
    & System.String^ ConfigurationName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigurationName

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ShowConfiguration](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

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ShowConfiguration2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ShowConfiguration2 Method (IModelDoc2)

Shows the named configuration by switching to that configuration and making it the active configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function ShowConfiguration2( _  
    ByVal ConfigurationName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ConfigurationName As System.String
```

```
Dim value As System.Boolean

value = instance.ShowConfiguration2(ConfigurationName)
```

C#

```
System.bool ShowConfiguration2(
    System.string ConfigurationName
)
```

C++/CLI

```
System.bool ShowConfiguration2(
    & System.String^ ConfigurationName
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigurationName

Name of the configuration to display

Return Value

True if the configuration is activated successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ShowConfiguration2](#).

Example

[Get Custom Property Values on Cut List Folders \(VBA\)](#)
[Iterate Through All Configurations \(VBA\)](#)
[Rebuild All Configurations \(VBA\)](#)
[Work With Configurations \(VBA\)](#)
[Save Configuration Data \(C#\)](#)
[Save Configuration Data \(VB.NET\)](#)
[Save Configuration Data \(VBA\)](#)

Remarks

Configurations allow you to save certain display characteristics with each of the assembly components. This method allows you to retrieve a previously saved configuration.

If you made changes to the model and the configuration to which you are switching needs to be updated to reflect those changes, then you must rebuild the model using [IModelDoc2::EditRebuild3](#). Otherwise, the FeatureManager design tree will show the model as needing to be rebuilt.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::EditConfiguration3 Method](#)
[IModelDoc2::AddConfiguration3 Method](#)
[IConfigurationManager::AddConfiguration Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ShowCosmeticThread Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ShowCosmeticThread Method (IModelDoc2)

Shows the selected cosmetic thread.

.NET Syntax

Visual Basic (Declaration)

```
Sub ShowCosmeticThread()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ShowCosmeticThread()
```

C#

```
void ShowCosmeticThread()
```

C++/CLI

```
void ShowCosmeticThread();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ShowCosmeticThread](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::HideCosmeticThread Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ShowFeatureDimensions Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ShowFeatureDimensions Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::GetUserPreferenceToggle](#) and [IModelDoc2::SetUserPreferenceToggle](#) and [swDisplayFeatureDimensions](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ShowFeatureDimensions()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ShowFeatureDimensions()
```

C#

```
void ShowFeatureDimensions()
```

C++/CLI

```
void ShowFeatureDimensions();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ShowFeatureDimensions](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ShowFeatureErrorDialog Property (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ShowFeatureErrorDialog Property (IModelDoc2)

Gets or sets whether to display the feature error dialog.

.NET Syntax

Visual Basic (Declaration)

Property ShowFeatureErrorDialog As System.Boolean

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Boolean

instance.ShowFeatureErrorDialog = value

value = instance.ShowFeatureErrorDialog
```

C#

```
System.bool ShowFeatureErrorDialog {get; set;}
```

C++/CLI

```
property System.bool ShowFeatureErrorDialog {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True displays an error dialog, false does not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ShowFeatureErrorDialog](#).

Remarks

This property controls the display of error dialogs generated when an error occurs during a rebuild of a specific feature in the current SOLIDWORKS Design session. Use [ISldWorks::GetUserPreferenceToggle](#) or [ISldWorks::SetUserPreferenceToggle](#) to get or set swShowErrorsEveryRebuild, which controls the display of error dialogs generated when a rebuild warning error message is generated.

You can wrap the rebuild API call with the following code to suppress rebuild warning dialogs:

```
bValue = swApp.GetUserPreferenceToggle(swShowErrorsEveryRebuild)
swApp.SetUserPreferenceToggle swShowErrorsEveryRebuild, false
swModel.ForceRebuild3
swApp.SetUserPreferenceToggle swShowErrorsEveryRebuild, bValue
```

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ForceRebuild3 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ShowNamedView Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ShowNamedView Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ShowNameView2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ShowNamedView( _  
    ByVal VName As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim VName As System.String  
  
instance.ShowNamedView(VName)
```

C#

```
void ShowNamedView(  
    System.string VName  
)
```

C++/CLI

```
void ShowNamedView(  
& System.String^ VName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VName

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ShowNamedView](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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ShowNamedView2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ShowNamedView2 Method (IModelDoc2)

Shows the specified view.

.NET Syntax

Visual Basic (Declaration)

```
Sub ShowNamedView2( _  
    ByVal VName As System.String, _
```

```
ByVal ViewId As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim VName As System.String  
Dim ViewId As System.Integer  
  
instance.ShowNamedView2(VName, ViewId)
```

C#

```
void ShowNamedView2(  
    System.string VName,  
    System.int ViewId  
)
```

C++/CLI

```
void ShowNamedView2(  
& System.String^ VName,  
& System.int ViewId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VName

Name of the view to display or an empty string to use ViewId instead

ViewId

ID of the view to display as defined by [swStandardViews_e](#) or -1 to use the VName argument instead; if you specify both VName and ViewId, then ViewId takes precedence if the two arguments do not resolve to the same view

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ShowNamedView2](#).

Example

[Add Component and Mate \(C++\)](#)
[Change to Isometric and Zoom to Fit View Mode \(VBA\)](#)
[Create Revolve Features \(VBA\)](#)
[Get Sketch Points in Wizard Hole \(VBA\)](#)
[Show Named View \(VBA\)](#)
[Add Spring to Motion Study \(C#\)](#)
[Add Spring to Motion Study \(VB.NET\)](#)
[Add Spring to Motion Study \(VBA\)](#)

Remarks

To set a named view to the active view in a drawing, the drawing view that contains the named view must be selected.

To orient the selected face to its Normal To view, specify ***Normal To** for VName and **-1** for ViewId.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::NameView Method](#)
[IModelDoc2::DeleteNamedView Method](#)
[IModelDocExtension::GetNamedViewRotation Method](#)
[IModelDocExtension::IGetNamedViewRotation Method](#)
[IModelDocExtension::ResetStandardViews Method \(\)](#)
[IModelDocExtension::UpdateStandardViews Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ShowSolidBody Method (IModelDoc2)

Shows the selected solid body.

.NET Syntax

Visual Basic (Declaration)

```
Sub ShowSolidBody()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ShowSolidBody()
```

C#

```
void ShowSolidBody()
```

C++/CLI

```
void ShowSolidBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ShowSolidBody](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::HideSolidBody Method](#)
[IModelDoc2::HideShowBodies Method](#)
[IFeatureManager::ShowBodies Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SimpleHole Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SimpleHole Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::SimpleHole](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SimpleHole(  
    ByVal Sd As System.Boolean, _  
    ByVal Flip As System.Boolean, _  
    ByVal Dir As System.Boolean, _  
    ByVal T1 As System.Integer, _  
    ByVal T2 As System.Integer, _  
    ByVal D1 As System.Double, _  
    ByVal D2 As System.Double, _  
    ByVal Dchk1 As System.Boolean, _  
    ByVal Dchk2 As System.Boolean, _  
    ByVal Ddir1 As System.Boolean, _  
    ByVal Ddir2 As System.Boolean, _  
    ByVal Dang1 As System.Double, _  
    ByVal Dang2 As System.Double, _
```

```

        ByVal OffsetReverse1 As System.Boolean, _
        ByVal OffsetReverse2 As System.Boolean _
    )

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Sd As System.Boolean
Dim Flip As System.Boolean
Dim Dir As System.Boolean
Dim T1 As System.Integer
Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean

instance.SimpleHole(Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1, Dang2,
OffsetReverse1, OffsetReverse2)

```

C#

```

void SimpleHole(
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2
)

```

C++/CLI

```

void SimpleHole(
    & System.bool Sd,
    & System.bool Flip,
    & System.bool Dir,
    & System.int T1,
    & System.int T2,
    & System.double D1,
    & System.double D2,
    & System.bool Dchk1,
    & System.bool Dchk2,
    & System.bool Ddir1,
    & System.bool Ddir2,
    & System.double Dang1,
    & System.double Dang2,
    & System.bool OffsetReverse1,
    & System.bool OffsetReverse2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sd

Flip
Dir
T1
T2
D1
D2
Dchk1
Dchk2
Ddir1
Ddir2
Dang1
Dang2
OffsetReverse1
OffsetReverse2

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SimpleHole](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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SimpleHole2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SimpleHole2 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::SimpleHole](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SimpleHole2( _  
    ByVal Dia As System.Double, _  
    ByVal Sd As System.Boolean, _  
    ByVal Flip As System.Boolean, _  
    ByVal Dir As System.Boolean, _  
    ByVal T1 As System.Integer, _  
    ByVal T2 As System.Integer, _  
    ByVal D1 As System.Double, _  
    ByVal D2 As System.Double, _  
    ByVal Dchk1 As System.Boolean, _  
    ByVal Dchk2 As System.Boolean, _  
    ByVal Ddir1 As System.Boolean, _  
    ByVal Ddir2 As System.Boolean, _  
    ByVal Dang1 As System.Double, _  
    ByVal Dang2 As System.Double, _  
    ByVal OffsetReverse1 As System.Boolean, _  
    ByVal OffsetReverse2 As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Dia As System.Double  
Dim Sd As System.Boolean  
Dim Flip As System.Boolean  
Dim Dir As System.Boolean  
Dim T1 As System.Integer
```

```

Dim T2 As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim Dchk1 As System.Boolean
Dim Dchk2 As System.Boolean
Dim Ddir1 As System.Boolean
Dim Ddir2 As System.Boolean
Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean

```

```

instance.SimpleHole2(Dia, Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1, Dang2,
OffsetReverse1, OffsetReverse2)

```

C#

```

void SimpleHole2(
    System.double Dia,
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2
)

```

C++/CLI

```

void SimpleHole2(
& System.double Dia,
& System.bool Sd,
& System.bool Flip,
& System.bool Dir,
& System.int T1,
& System.int T2,
& System.double D1,
& System.double D2,
& System.bool Dchk1,
& System.bool Dchk2,
& System.bool Ddir1,
& System.bool Ddir2,
& System.double Dang1,
& System.double Dang2,
& System.bool OffsetReverse1,
& System.bool OffsetReverse2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Dia

Sd

Flip

Dir

T1

T2

D1

D2

Dchk1
Dchk2
Ddir1
Ddir2
Dang1
Dang2
OffsetReverse1
OffsetReverse2

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SimpleHole2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SimpleHole3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SimpleHole3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::SimpleHole](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SimpleHole3( _  
    ByVal Dia As System.Double, _  
    ByVal Sd As System.Boolean, _  
    ByVal Flip As System.Boolean, _  
    ByVal Dir As System.Boolean, _  
    ByVal T1 As System.Integer, _  
    ByVal T2 As System.Integer, _  
    ByVal D1 As System.Double, _  
    ByVal D2 As System.Double, _  
    ByVal Dchk1 As System.Boolean, _  
    ByVal Dchk2 As System.Boolean, _  
    ByVal Ddir1 As System.Boolean, _  
    ByVal Ddir2 As System.Boolean, _  
    ByVal Dang1 As System.Double, _  
    ByVal Dang2 As System.Double, _  
    ByVal OffsetReverse1 As System.Boolean, _  
    ByVal OffsetReverse2 As System.Boolean, _  
    ByVal TranslateSurface1 As System.Boolean, _  
    ByVal TranslateSurface2 As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Dia As System.Double  
Dim Sd As System.Boolean  
Dim Flip As System.Boolean  
Dim Dir As System.Boolean  
Dim T1 As System.Integer  
Dim T2 As System.Integer  
Dim D1 As System.Double  
Dim D2 As System.Double  
Dim Dchk1 As System.Boolean  
Dim Dchk2 As System.Boolean  
Dim Ddir1 As System.Boolean  
Dim Ddir2 As System.Boolean
```

```

Dim Dang1 As System.Double
Dim Dang2 As System.Double
Dim OffsetReverse1 As System.Boolean
Dim OffsetReverse2 As System.Boolean
Dim TranslateSurface1 As System.Boolean
Dim TranslateSurface2 As System.Boolean

```

```

instance.SimpleHole3(Dia, Sd, Flip, Dir, T1, T2, D1, D2, Dchk1, Dchk2, Ddir1, Ddir2, Dang1, Dang2,
OffsetReverse1, OffsetReverse2, TranslateSurface1, TranslateSurface2)

```

C#

```

void SimpleHole3(
    System.double Dia,
    System.bool Sd,
    System.bool Flip,
    System.bool Dir,
    System.int T1,
    System.int T2,
    System.double D1,
    System.double D2,
    System.bool Dchk1,
    System.bool Dchk2,
    System.bool Ddir1,
    System.bool Ddir2,
    System.double Dang1,
    System.double Dang2,
    System.bool OffsetReverse1,
    System.bool OffsetReverse2,
    System.bool TranslateSurface1,
    System.bool TranslateSurface2
)

```

C++/CLI

```

void SimpleHole3(
    & System.double Dia,
    & System.bool Sd,
    & System.bool Flip,
    & System.bool Dir,
    & System.int T1,
    & System.int T2,
    & System.double D1,
    & System.double D2,
    & System.bool Dchk1,
    & System.bool Dchk2,
    & System.bool Ddir1,
    & System.bool Ddir2,
    & System.double Dang1,
    & System.double Dang2,
    & System.bool OffsetReverse1,
    & System.bool OffsetReverse2,
    & System.bool TranslateSurface1,
    & System.bool TranslateSurface2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Dia

Sd

Flip

Dir

T1

T2

D1

D2

Dchk1

Dchk2
Ddir1
Ddir2
Dang1
Dang2
OffsetReverse1
OffsetReverse2
TranslateSurface1
TranslateSurface2

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SimpleHole3](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SimplifySpline Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SimplifySpline Method (IModelDoc2)

Obsolete. Superseded by [ISketchSpline::Simplify](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SimplifySpline( _  
    ByVal ToleranceIn As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ToleranceIn As System.Double  
  
instance.SimplifySpline(ToleranceIn)
```

C#

```
void SimplifySpline(  
    System.double ToleranceIn  
)
```

C++/CLI

```
void SimplifySpline(  
& System.double ToleranceIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToleranceIn

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SimplifySpline](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Sketch3DIntersections Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : Sketch3DIntersections Method (IModelDoc2)

Creates new sketch segments based on the selected surfaces.

.NET Syntax

Visual Basic (Declaration)

```
Sub Sketch3DIntersections()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.Sketch3DIntersections()
```

C#

```
void Sketch3DIntersections()
```

C++/CLI

```
void Sketch3DIntersections();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::Sketch3DIntersections](#).

Remarks

The new sketch segments are added either to the active sketch or to an appropriate new sketch.

If the active sketch is a 2D sketch and the intersection curves are not in that plane, then the resulting sketch segments are projected onto the plane of the sketch.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISurface::GetIntersectCurveCount2 Method](#)
[ISurface::IntersectCurve2 Method](#)
[ISurface::IntersectCurve2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchAddConstraints Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchAddConstraints Method (IModelDoc2)

Adds the specified constraint to the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchAddConstraints( _
    ByVal IdStr As System.String _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim IdStr As System.String

instance.SketchAddConstraints(IdStr)
```

C#

```
void SketchAddConstraints(
    System.string IdStr
)
```

C++/CLI

```
void SketchAddConstraints(
    & System.String^ IdStr
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IdStr

Constraint (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchAddConstraints](#).

Example

[Constrain Sketch \(VBA\)](#)
[Constrain Sketch \(VB.NET\)](#)
[Constrain Sketch \(C#\)](#)

Remarks

SOLIDWORKS 2005 and earlier

- . sgHORIZONTAL (2D sketch)
- . sgHORIZONTAL (3D sketch)
- . sgHORIZPOINTS (2D sketch)
- . sgHORIZPOINTS (3D sketch)
- . sgVERTICAL (2D sketch)
- . sgVERTICAL (3D sketch)
- . sgVERTPOINTS (2D sketch)
- . sgVERTPOINTS (3D sketch)
- . sgALONGZPOINTS
- . sgALONGZ
- . sgCOLINEAR
- . sgCORADIAL
- . sgPERPENDICULAR
- . sgPARALLEL
- . sgTANGENT
- . sgCONCENTRIC
- . sgCOINCIDENT

SOLIDWORKS 2006 and later

- . sgHORIZONTAL2D
- . sgALONGX3D
- . sgHORIZONTALPOINTS2D
- . sgALONGXPOINTS3D
- . sgVERTICAL2D
- . sgALONGY3D
- . sgVERTICALPOINTS2D
- . sgALONGYPOINTS3D
- . sgALONGZPOINTS3D
- . sgALONGZ3D
- . sgCOLINEAR
- . sgCORADIAL
- . sgPERPENDICULAR
- . sgPARALLEL
- . sgTANGENT
- . sgCONCENTRIC
- . sgCOINCIDENT

- . sgSYMMETRIC
 - . sgATMIDDLE
 - . sgATINTERSECT
 - . sgATPIERCE
 - . sgFIXED
 - . sgANGLE
 - . sgARCANG180
 - . sgARCANG270
 - . sgARCANG90
 - . sgARCANGBOTTOM
 - . sgARCANGLEFT
 - . sgARCANGRIGHT
 - . sgARCANGTOP
 - . sgDIAMETER
 - . sgDISTANCE
 - . sgSAMELENGTH
 - . sgOFFSETEDGE
 - . sgSNAPANGLE
 - . sgSNAPGRID
 - . sgSNAPLENGTH
 - . sgUSEEDGE

- . sgSYMMETRIC
 - . sgATMIDDLE
 - . sgATINTERSECT
 - . sgATPIERCE
 - . sgFIXED
 - . sgANGLE
 - . sgARCANG180
 - . sgARCANG270
 - . sgARCANG90
 - . sgARCANGBOTTOM
 - . sgARCANGLEFT
 - . sgARCANGRIGHT
 - . sgARCANGTOP
 - . sgDIAMETER
 - . sgDISTANCE
 - . sgSAMELENGTH
 - . sgOFFSETEDGE
 - . sgSNAPANGLE
 - . sgSNAPGRID
 - . sgSNAPLENGTH
 - . sgUSEEDGE
 - . sgMERGEPOINTS

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchConstrainCoincident Method](#)
[IModelDoc2::SketchConstrainConcentric Method](#)
[IModelDoc2::SketchConstrainParallel Method](#)
[IModelDoc2::SketchConstrainPerp Method](#)
[IModelDoc2::SketchConstrainTangent Method](#)
[IModelDoc2::SketchConstraintsDelAll Method](#)
[IModelDoc2::SketchConstraintsDelAll Method](#)
[IModelDoc2::ViewConstraint Method](#)
[IModelDoc2::SkToolsAutoConstr Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchAlign Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchAlign Method (IModelDoc2)

Aligns the selected sketch entities.

 I.NET Syntax

Visual Basic (Declaration)

Sub SketchAlign()

Visual Basic (Usage)



Dim instance As [IModelDoc2](#)

instance.SketchAlign()

C#

```
void SketchAlign()
```

C++/CLI

```
void SketchAlign();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchAlign](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchArc Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchArc Method (IModelDoc2)

Creates an arc in the current model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchArc()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.SketchArc()
```

C#

```
void SketchArc()
```

C++/CLI

```
void SketchArc();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchArc](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISketchArc Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchCenterline Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchCenterline Method (IModelDoc2)

Adds a centerline to the current model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchCenterline()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.SketchCenterline()
```

C#

```
void SketchCenterline()
```

C++/CLI

```
void SketchCenterline();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchCenterline](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ICenterLine Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchChamfer Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchChamfer Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreateChamfer](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchChamfer(  
    ByVal AngleORdist As System.Double, _  
    ByVal Dist1 As System.Double, _  
    ByVal Options As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim AngleORdist As System.Double  
Dim Dist1 As System.Double  
Dim Options As System.Integer
```

```
instance.SketchChamfer(AngleORdist, Dist1, Options)
```

C#

```
void SketchChamfer(  
    System.double AngleORdist,  
    System.double Dist1,  
    System.int Options  
)
```

C++/CLI

```
void SketchChamfer(  
& System.double AngleORdist,  
& System.double Dist1,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AngleORdist

Angle of the chamfer if using the Angle-Distance option or the distance of the second distance if using the Distance-Distance option

Dist1

Distance of the chamfer

Options

0 = Angle - Distance Chamfer

1 = Distance - Distance Chamfer

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchChamfer](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchCircle Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchCircle Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateCircle2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchCircle()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.SketchCircle()
```

C#

```
void SketchCircle()
```

C++/CLI

```
void SketchCircle();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchCircle](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchConstrainCoincident Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchConstrainCoincident Method (IModelDoc2)

Makes the selected sketch entities coincident.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchConstrainCoincident()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.SketchConstrainCoincident()
```

C#

```
void SketchConstrainCoincident()
```

C++/CLI

```
void SketchConstrainCoincident();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchConstrainCoincident](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchAddConstraints Method](#)
[IModelDoc2::SketchConstrainConcentric Method](#)
[IModelDoc2::SketchConstrainParallel Method](#)
[IModelDoc2::SketchConstrainPerp Method](#)
[IModelDoc2::SketchConstrainTangent Method](#)
[IModelDoc2::SketchConstraintsDel Method](#)
[IModelDoc2::SketchConstraintsDelAll Method](#)
[IModelDoc2::SkToolsAutoConstr Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchConstrainConcentric Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchConstrainConcentric Method (IModelDoc2)

Makes the selected sketch entities concentric.

 **.NET Syntax**

Visual Basic (Declaration)

Sub SketchConstrainConcentric()

Visual Basic (Usage)  Copy Code

Dim instance As [IModelDoc2](#)

instance.SketchConstrainConcentric()

C#

void SketchConstrainConcentric()

C++/CLI

void SketchConstrainConcentric();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 **Visual Basic for Applications (VBA) Syntax**

See [ModelDoc2::SketchConstrainConcentric](#).

 **See Also**

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchAddConstraints Method](#)
[IModelDoc2::SketchConstrainCoincident Method](#)
[IModelDoc2::SketchConstrainParallel Method](#)
[IModelDoc2::SketchConstrainPerp Method](#)
[IModelDoc2::SketchConstrainTangent Method](#)
[IModelDoc2::SketchConstraintsDel Method](#)
[IModelDoc2::SketchConstraintsDelAll Method](#)
[IModelDoc2::SkToolsAutoConstr Method](#)

 **Availability**

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help [Send comments on this topic.](#)

SketchConstrainParallel Method (IModelDoc2)

[See Also](#)

 

[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchConstrainParallel Method (IModelDoc2)

Makes the selected sketch entities parallel.

 **.NET Syntax**

Visual Basic (Declaration)

Sub SketchConstrainParallel()

Visual Basic (Usage)  Copy Code

Dim instance As [IModelDoc2](#)

instance.SketchConstrainParallel()

C#

void SketchConstrainParallel()

C++/CLI

```
void SketchConstrainParallel();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchConstrainParallel](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchAddConstraints Method](#)
[IModelDoc2::SketchConstrainCoincident Method](#)
[IModelDoc2::SketchConstrainConcentric Method](#)
[IModelDoc2::SketchConstrainPerp Method](#)
[IModelDoc2::SketchConstrainTangent Method](#)
[IModelDoc2::SketchConstraintsDel Method](#)
[IModelDoc2::SketchConstraintsDelAll Method](#)
[IModelDoc2::SkToolsAutoConstr Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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SketchConstrainPerp Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchConstrainPerp Method (IModelDoc2)

Makes the selected sketch entities perpendicular.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchConstrainPerp()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.SketchConstrainPerp()
```

C#

```
void SketchConstrainPerp()
```

C++/CLI

```
void SketchConstrainPerp();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchConstrainPerp](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchAddConstraints Method](#)
[IModelDoc2::SketchConstrainCoincident Method](#)
[IModelDoc2::SketchConstrainConcentric Method](#)
[IModelDoc2::SketchConstrainParallel Method](#)
[IModelDoc2::SketchConstrainTangent Method](#)
[IModelDoc2::SketchConstraintsDel Method](#)
[IModelDoc2::SketchConstraintsDelAll Method](#)
[IModelDoc2::SkToolsAutoConstr Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchConstrainTangent Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchConstrainTangent Method (IModelDoc2)

Makes the selected entities tangent.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchConstrainTangent()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.SketchConstrainTangent()
```

C#

```
void SketchConstrainTangent()
```

C++/CLI

```
void SketchConstrainTangent();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchConstrainTangent](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchAddConstraints Method](#)
[IModelDoc2::SketchConstrainCoincident Method](#)
[IModelDoc2::SketchConstrainConcentric Method](#)
[IModelDoc2::SketchConstrainParallel Method](#)
[IModelDoc2::SketchConstrainPerp Method](#)
[IModelDoc2::SketchConstraintsDel Method](#)
[IModelDoc2::SketchConstraintsDelAll Method](#)
[IModelDoc2::SkToolsAutoConstr Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchConstraintsDel Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchConstraintsDel Method (IModelDoc2)

Deletes the specified relationship (constraint) on the currently selected sketch item.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchConstraintsDel( _
    ByVal ConstrInd As System.Integer, _
    ByVal IdStr As System.String _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim ConstrInd As System.Integer
Dim IdStr As System.String

instance.SketchConstraintsDel(ConstrInd, IdStr)
```

C#

```
void SketchConstraintsDel(
    System.int ConstrInd,
    System.string IdStr
)
```

C++/CLI

```
void SketchConstraintsDel(
    & System.int ConstrInd,
    & System.String^ IdStr
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConstrInd
Constraint number on the selected entity; 0-based index
IdStr
Constraint to delete (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchConstraintsDel](#).

Remarks

To delete a tangency relation, which is the third relation on the selected arc, specify:

Part.SketchConstraintsDel 2, "sgTANGENT"

The available constraints are as follows. Pass them as strings embedded in quotes:

SOLIDWORKS 2005 and earlier	SOLIDWORKS 2006 and later
- sgHORIZONTAL (2D sketch) - sgHORIZONTAL (3D sketch) - sgHORIZPOINTS (2D sketch) - sgHORIZPOINTS (3D sketch) - sgVERTICAL (2D sketch) - sgVERTICAL (3D sketch) - sgVERTPOINTS (2D sketch) - sgVERTPOINTS (3D sketch) - sgALONGZPOINTS - sgALONGZ - sgCOLINEAR - sgCORADIAL - sgPERPENDICULAR	- sgHORIZONTAL2D - sgALONGX3D - sgHORIZONTALPOINTS2D - sgALONGXPOINTS3D - sgVERTICAL2D - sgALONGY3D - sgVERTICALPOINTS2D - sgALONGYPOINTS3D - sgALONGZPOINTS3D - sgALONGZ3D - sgCOLINEAR - sgCORADIAL - sgPERPENDICULAR

- | | |
|--|---|
| <ul style="list-style-type: none"> . sgPARALLEL . sgTANGENT . sgCONCENTRIC . sgCOINCIDENT . sgSYMMETRIC . sgATMIDDLE . sgATINTERSECT . sgATPIERCE . sgFIXED . sgANGLE . sgARCANG180 . sgARCANG270 . sgARCANG90 . sgARCANGBOTTOM . sgARCANGLEFT . sgARCANGRIGHT . sgARCANGTOP . sgDIAMETER . sgDISTANCE . sgSAMELENGTH . sgOFFSETEGE . sgSNAPANGLE . sgSNAPGRID . sgSNAPLENGTH . sgUSEEDGE | <ul style="list-style-type: none"> . sgPARALLEL . sgTANGENT . sgCONCENTRIC . sgCOINCIDENT . sgSYMMETRIC . sgATMIDDLE . sgATINTERSECT . sgATPIERCE . sgFIXED . sgANGLE . sgARCANG180 . sgARCANG270 . sgARCANG90 . sgARCANGBOTTOM . sgARCANGLEFT . sgARCANGRIGHT . sgARCANGTOP . sgDIAMETER . sgDISTANCE . sgSAMELENGTH . sgOFFSETEGE . sgSNAPANGLE . sgSNAPGRID . sgSNAPLENGTH . sgUSEEDGE . sgMERGEPOINTS |
|--|---|

NOTE: Although SOLIDWORKS 2006 and later accepts the constraints shown in the first column, you are encouraged to use the constraints shown in the second column.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchAddConstraints Method](#)
[IModelDoc2::SketchConstrainCoincident Method](#)
[IModelDoc2::SketchConstrainConcentric Method](#)
[IModelDoc2::SketchConstrainParallel Method](#)
[IModelDoc2::SketchConstrainPerp Method](#)
[IModelDoc2::SketchConstrainTangent Method](#)
[IModelDoc2::SketchConstraintsDelAll Method](#)
[IModelDoc2::SkToolsAutoConstr Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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SketchConstraintsDelAll Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchConstraintsDelAll Method (IModelDoc2)

Deletes all of the constraints on the currently selected sketch segment.

.NET Syntax

Visual Basic (Declaration)

Sub SketchConstraintsDelAll()

```
Dim instance As IModelDoc2

instance.SketchConstraintsDelAll()
```

C#

```
void SketchConstraintsDelAll()
```

C++/CLI

```
void SketchConstraintsDelAll();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchConstraintsDelAll](#).

Example

[Delete All Constraints in Selected Sketch \(VBA\)](#)

Remarks

Only the constraints on the currently selected sketch segment are deleted. If two or more sketch segments are selected, this method has no effect.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchAddConstraints Method](#)
[IModelDoc2::SketchConstrainCoincident Method](#)
[IModelDoc2::SketchConstrainConcentric Method](#)
[IModelDoc2::SketchConstrainParallel Method](#)
[IModelDoc2::SketchConstrainPerp Method](#)
[IModelDoc2::SketchConstraintsDel Method](#)
[IModelDoc2::SketchConstrainTangent Method](#)
[IModelDoc2::SkToolsAutoConstr Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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SketchConvertIsoCurves Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchConvertIsoCurves Method (IModelDoc2)

Converts ISO-parametric curves on a selected surface into a sketch entity.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchConvertIsoCurves( _
    ByVal PercentRatio As System.Double, _
    ByVal VORuDir As System.Boolean, _
    ByVal DoConstrain As System.Boolean, _
    ByVal SkipHoles As System.Boolean _
)
```

Visual Basic (Usage)

Copy Code

```
Dim instance As IModelDoc2
Dim PercentRatio As System.Double
```

```
Dim VORuDir As System.Boolean
Dim DoConstrain As System.Boolean
Dim SkipHoles As System.Boolean
```

```
instance.SketchConvertIsoCurves(PercentRatio, VORuDir, DoConstrain, SkipHoles)
```

C#

```
void SketchConvertIsoCurves(
    System.double PercentRatio,
    System.bool VORuDir,
    System.bool DoConstrain,
    System.bool SkipHoles
)
```

C++/CLI

```
void SketchConvertIsoCurves(
    & System.double PercentRatio,
    & System.bool VORuDir,
    & System.bool DoConstrain,
    & System.bool SkipHoles
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PercentRatio
Value for percent ratio

VORuDir
True for V direction, false for UDirection

DoConstrain
True if you want to constrain these new sketch entities, false if not

SkipHoles
True if you want to skip the holes in this surface, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchConvertIsoCurves](#).

Example

[Divide Surface into 3D Sketches \(VBA\)](#)
[Convert Curves into 3D Sketches \(C#\)](#)
[Convert Curves into 3D Sketches \(VB.NET\)](#)
[Convert Curves into 3D Sketches \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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SketchFillet Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchFillet Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::SketchFillet2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchFillet( _  
    ByVal Rad As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Rad As System.Double  
  
instance.SketchFillet(Rad)
```

C#

```
void SketchFillet(  
    System.double Rad  
)
```

C++/CLI

```
void SketchFillet(  
&    System.double Rad  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Rad

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchFillet](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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SketchFillet1 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchFillet1 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::SketchFillet2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchFillet1( _  
    ByVal Rad As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Rad As System.Double  
  
instance.SketchFillet1(Rad)
```

C#

```
void SketchFillet1(  
    System.double Rad  
)
```

C++/CLI

```
void SketchFillet1(
& System.double Rad
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Rad

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchFillet1](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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SketchFillet2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchFillet2 Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreateFillet](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SketchFillet2( _
    ByVal Rad As System.Double, _
    ByVal ConstrainedCorners As System.Short _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Rad As System.Double
Dim ConstrainedCorners As System.Short
Dim value As System.Boolean

value = instance.SketchFillet2(Rad, ConstrainedCorners)
```

C#

```
System.bool SketchFillet2(
    System.double Rad,
    System.short ConstrainedCorners
)
```

C++/CLI

```
System.bool SketchFillet2(
& System.double Rad,
& System.short ConstrainedCorners
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Rad

Radius of the fillet in meters

ConstrainedCorners

Action to take if the corner to be filleted is constrained or has a dimension (see **Remarks**)

Return Value

True if the fillet is created, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchFillet2](#).

Remarks

The ConstrainedCorners argument:

- Indicates what action to take if the corner to be filleted is constrained in some manner or has a dimension related to it. In this case, adding a fillet to the corner cannot be done without certain consequences. If the corner is not involved with any constraints, this argument is ignored.
- Can take one of the values found in [swConstrainedCornerAction](#) e.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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SketchManager Property (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchManager Property (IModelDoc2)

Gets the sketch manager, which allows access to sketch-creation routines.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property SketchManager As SketchManager

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As SketchManager

value = instance.SketchManager
```

C#

```
SketchManager SketchManager {get;}
```

C++/CLI

```
property SketchManager^ SketchManager {
    SketchManager^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Sketch manager](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchManager](#).

Example

[Dynamically Mirror Sketch Entities \(VBA\)](#)
[Get Sketch Entities \(VBA\)](#)
[Offset Selected Edges in Active Sketch \(VBA\)](#)

[Insert and Resize Sketch Slot \(C#\)](#)
[Insert and Resize Sketch Slot \(VB.NET\)](#)
[Insert and Resize Sketch Slot \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

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SketchMirror Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchMirror Method (IModelDoc2)

Creates new entities that are mirror images of the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchMirror()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.SketchMirror()
```

C#

```
void SketchMirror()
```

C++/CLI

```
void SketchMirror();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchMirror](#).

Remarks

If the entity to mirror is:

- a sketch segment, then use a selection mark of 1
- a centerline, then use a selection mark of 2

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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SketchModifyFlip Method (IModelDoc2)

[See Also](#) [Example](#)



Flips the active or selected sketch about the specified coordinate system axis.

 .NET Syntax

Visual Basic (Declaration)

```
Sub SketchModifyFlip( _  
    ByVal AxisFlag As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim AxisFlag As System.Integer  
  
instance.SketchModifyFlip(AxisFlag)
```

C#

```
void SketchModifyFlip(  
    System.int AxisFlag  
)
```

C++/CLI

```
void SketchModifyFlip(  
    & System.int AxisFlag  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AxisFlag

Axis about which to flip the sketch:

- X = 1
- Y = 2

 Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchModifyFlip](#).

 Example

[Flip Sketch \(VBA\)](#)
[Flip Sketch \(VB.NET\)](#)
[Flip Sketch \(C#\)](#)

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SketchModifyRotate Method (IModelDoc2)

[See Also](#) [Example](#)



Rotates the coordinate system of the active or selected sketch.

 .NET Syntax

Visual Basic (Declaration)

```
Sub SketchModifyRotate( _
    ByVal CenterX As System.Double, _
    ByVal CenterY As System.Double, _
    ByVal Angle As System.Double _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim CenterX As System.Double
Dim CenterY As System.Double
Dim Angle As System.Double

instance.SketchModifyRotate(CenterX, CenterY, Angle)
```

C#

```
void SketchModifyRotate(
    System.double CenterX,
    System.double CenterY,
    System.double Angle
)
```

C++/CLI

```
void SketchModifyRotate(
    & System.double CenterX,
    & System.double CenterY,
    & System.double Angle
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CenterX
X point to rotate about

CenterY
Y point to rotate about

Angle
Angle of rotation

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchModifyRotate](#).

Example

[Rotate, Scale, and Copy Sketch \(C#\)](#)
[Rotate, Scale, and Copy Sketch \(VB.NET\)](#)
[Rotate, Scale, and Copy Sketch \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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SketchModifyScale Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchModifyScale Method (IModelDoc2)

Scales the active or selected sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function SketchModifyScale( _  
    ByVal ScaleFactor As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ScaleFactor As System.Double  
Dim value As System.Boolean  
  
value = instance.SketchModifyScale(ScaleFactor)
```

C#

```
System.bool SketchModifyScale(  
    System.double ScaleFactor  
)
```

C++/CLI

```
System.bool SketchModifyScale(  
&    System.double ScaleFactor  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ScaleFactor

Amount by which to scale sketch

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchModifyScale](#).

Example

[Rotate, Scale, and Copy Sketch \(C#\)](#)
[Rotate, Scale, and Copy Sketch \(VB.NET\)](#)
[Rotate, Scale, and Copy Sketch \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ToolsSketchScale Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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SketchModifyTranslate Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchModifyTranslate Method (IModelDoc2)

Translates the coordinate system of the active or selected sketch.

.NET Syntax

Visual Basic (Declaration)

```

Sub SketchModifyTranslate( _
    ByVal StartX As System.Double, _
    ByVal StartY As System.Double, _
    ByVal EndX As System.Double, _
    ByVal EndY As System.Double _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim StartX As System.Double
Dim StartY As System.Double
Dim EndX As System.Double
Dim EndY As System.Double

```

```
instance.SketchModifyTranslate(StartX, StartY, EndX, EndY)
```

C#

```

void SketchModifyTranslate(
    System.double StartX,
    System.double StartY,
    System.double EndX,
    System.double EndY
)

```

C++/CLI

```

void SketchModifyTranslate(
    & System.double StartX,
    & System.double StartY,
    & System.double EndX,
    & System.double EndY
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StartX

X sketch value defining the from-position

StartY

Y sketch value defining the from-position

EndX

X sketch value defining the to-position

EndY

Y sketch value defining the to-position

Return Value

The sketch is translated from the XY start point position to the XY end point position.

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchModifyTranslate](#).

Example

[Translate Sketch \(VBA\)](#)
[Translate Sketch \(VB.NET\)](#)
[Translate Sketch \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ToolsSketchTranslate Method](#)

Availability

SketchOffset Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchOffset Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::SketchOffset2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchOffset( _  
    ByVal Offset As System.Double, _  
    ByVal ContourMode As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Offset As System.Double  
Dim ContourMode As System.Boolean  
  
instance.SketchOffset(Offset, ContourMode)
```

C#

```
void SketchOffset(  
    System.double Offset,  
    System.bool ContourMode  
)
```

C++/CLI

```
void SketchOffset(  
& System.double Offset,  
& System.bool ContourMode  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Offset

ContourMode

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchOffset](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SketchOffset2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchOffset2 Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::SketchOffset](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SketchOffset2( _
    ByVal Offset As System.Double, _
    ByVal BothDirections As System.Boolean, _
    ByVal Chain As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Offset As System.Double
Dim BothDirections As System.Boolean
Dim Chain As System.Boolean
Dim value As System.Boolean

value = instance.SketchOffset2(Offset, BothDirections, Chain)
```

C#

```
System.bool SketchOffset2(
    System.double Offset,
    System.bool BothDirections,
    System.bool Chain
)
```

C++/CLI

```
System.bool SketchOffset2(
    & System.double Offset,
    & System.bool BothDirections,
    & System.bool Chain
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Offset

Offset value; negative value offsets in opposite direction

BothDirections

True to offset in both directions, false to not

Chain

True if you want the entire chain of entities offset, false if you want only the selected sketch entities offset (see **Remarks**)

Return Value

True if the offset is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchOffset2](#).

Remarks

Specifying true for the Chain argument offsets the selected entity and any other entities that belong to the same contour or chain (contiguous, geometric entities like edges).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchOffsetEdges Method](#)
[IModelDoc2::SketchOffsetEntities2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SketchOffsetEdges Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchOffsetEdges Method (IModelDoc2)

Offsets the edges of the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchOffsetEdges( _  
    ByVal Val As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
  
instance.SketchOffsetEdges(Val)
```

C#

```
void SketchOffsetEdges(  
    System.double Val  
)
```

C++/CLI

```
void SketchOffsetEdges(  
& System.double Val  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val
Offset value in meters

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchOffsetEdges](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchOffset2 Method](#)
[IModelDoc2::SketchOffsetEntities2 Method](#)
[IModelDocExtension::SketchOffsetOnSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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SketchOffsetEntities Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchOffsetEntities Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::SketchOffsetEntities2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchOffsetEntities( _  
    ByVal Offset As System.Double, _
```

```
ByVal Flip As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Offset As System.Double  
Dim Flip As System.Boolean  
  
instance.SketchOffsetEntities(Offset, Flip)
```

C#

```
void SketchOffsetEntities(  
    System.double Offset,  
    System.bool Flip  
)
```

C++/CLI

```
void SketchOffsetEntities(  
& System.double Offset,  
& System.bool Flip  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Offset

Flip

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchOffsetEntities](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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SketchOffsetEntities2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchOffsetEntities2 Method (IModelDoc2)

Generates entities in the active sketch by offsetting the selected geometry by the specified amount.

.NET Syntax

Visual Basic (Declaration)

```
Function SketchOffsetEntities2( _  
    ByVal Offset As System.Double, _  
    ByVal BothDirections As System.Boolean, _  
    ByVal Chain As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Offset As System.Double  
Dim BothDirections As System.Boolean  
Dim Chain As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SketchOffsetEntities2(Offset, BothDirections, Chain)
```

C#

```
System.bool SketchOffsetEntities2(  
    System.double Offset,  
    System.bool BothDirections,  
    System.bool Chain  
)
```

C++/CLI

```
System.bool SketchOffsetEntities2(  
& System.double Offset,  
& System.bool BothDirections,  
& System.bool Chain  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Offset

Offset distance in meters

BothDirections

True to offset in both directions, false to not

Chain

True if you want entire chain of entities offset, false if you want only selected sketch entities offset (see **Remarks**)

Return Value

True if the offset is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchOffsetEntities2](#).

Remarks

The geometry selected for offset can be an edge, loop, face, external sketch curve, external sketch contour, set of edges, or set of external sketch curves.

Specifying true for the Chain argument offsets the selected entity and any other entities that belong to the same contour or chain (contiguous, geometric entities like edges).

NOTE: If the selected geometry is a sketch item, it must be an external sketch curve (for example, it cannot be an item in the active sketch). To offset sketch segments within the active sketch, use [IModelDoc2::SketchOffset2](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::SketchOffsetEdges Method](#)

[ISketchManager::SketchUseEdge2 Method](#)

[IModelDocExtension::SketchOffsetOnSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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SketchParabola Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchParabola Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreateParabola](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchParabola( _  
    ByVal Val1 As System.Double, _
```

```

ByVal Val2 As System.Double, _
ByVal Z1 As System.Double, _
ByVal Val3 As System.Double, _
ByVal Val4 As System.Double, _
ByVal Z2 As System.Double, _
ByVal Val5 As System.Double, _
ByVal Val6 As System.Double, _
ByVal Z3 As System.Double, _
ByVal Val7 As System.Double, _
ByVal Val8 As System.Double, _
ByVal Z4 As System.Double _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Val1 As System.Double
Dim Val2 As System.Double
Dim Z1 As System.Double
Dim Val3 As System.Double
Dim Val4 As System.Double
Dim Z2 As System.Double
Dim Val5 As System.Double
Dim Val6 As System.Double
Dim Z3 As System.Double
Dim Val7 As System.Double
Dim Val8 As System.Double
Dim Z4 As System.Double

```

```
instance.SketchParabola(Val1, Val2, Z1, Val3, Val4, Z2, Val5, Val6, Z3, Val7, Val8, Z4)
```

C#

```

void SketchParabola(
    System.double Val1,
    System.double Val2,
    System.double Z1,
    System.double Val3,
    System.double Val4,
    System.double Z2,
    System.double Val5,
    System.double Val6,
    System.double Z3,
    System.double Val7,
    System.double Val8,
    System.double Z4
)

```

C++/CLI

```

void SketchParabola(
    & System.double Val1,
    & System.double Val2,
    & System.double Z1,
    & System.double Val3,
    & System.double Val4,
    & System.double Z2,
    & System.double Val5,
    & System.double Val6,
    & System.double Z3,
    & System.double Val7,
    & System.double Val8,
    & System.double Z4
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[Val1](#)

X location of the focus of the parabola

Val2
Y location of the focus of the parabola

Z1
Z location of the focus of the parabola

Val3
X location of the apex of the parabola

Val4
Y location of the apex of the parabola

Z2
Z location of the apex of the parabola

Val5
X location of the start of the parabola

Val6
Y location of the start of the parabola

Z3
Z location of the start of the parabola

Val7
X location of the end of the parabola

Val8
Y location of the end of the parabola

Z4
Z location of the end of the parabola

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchParabola](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISketchParabola Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchPoint Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchPoint Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreatePoint2](#) and [IModelDoc2::ICreatePoint2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchPoint( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
  
instance.SketchPoint(X, Y, Z)
```

C#

```
void SketchPoint(  
    System.double X,
```



```
System.double Y,  
System.double Z  
)
```

C++/CLI

```
void SketchPoint(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchPoint](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchPolygon Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchPolygon Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreatePolygon](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SketchPolygon( _  
    ByVal XCenter As System.Double, _  
    ByVal YCenter As System.Double, _  
    ByVal XEdge As System.Double, _  
    ByVal YEdge As System.Double, _  
    ByVal NSides As System.Integer, _  
    ByVal BInscribed As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim XCenter As System.Double  
Dim YCenter As System.Double  
Dim XEdge As System.Double  
Dim YEdge As System.Double  
Dim NSides As System.Integer  
Dim BInscribed As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.SketchPolygon(XCenter, YCenter, XEdge, YEdge, NSides, BInscribed)
```

C#

```
System.bool SketchPolygon(  
    System.double XCenter,  
    System.double YCenter,  
    System.double XEdge,
```

```
System.double YEdge,  
System.int NSides,  
System.bool BInscribed  
)
```

C++/CLI

```
System.bool SketchPolygon(  
& System.double XCenter,  
& System.double YCenter,  
& System.double XEdge,  
& System.double YEdge,  
& System.int NSides,  
& System.bool BInscribed  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XCenter

X component of the polygon's center

YCenter

Y component of the polygon's center

XEdge

X component of the first vertex on the polygon

YEdge

Y component of the first vertex on the polygon

NSides

Number of sides for the polygon

BInscribed

True to show an inscribed construction circle, false to show a circumscribed construction circle

Return Value

True if polygon created, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchPolygon](#).

Remarks

After using this method, the PropertyManager page is in edit mode for the polygon. To exit this PropertyManager page and complete the operation, use [IModelDoc2::SetPickMode](#), [IModelDoc2::ClearSelection2](#), or exit the sketch.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchRectangle Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchRectangle Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreateCornerRectangle](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchRectangle( _  
    ByVal Val1 As System.Double, _
```

```

    ByVal Val2 As System.Double, _
    ByVal Z1 As System.Double, _
    ByVal Val3 As System.Double, _
    ByVal Val4 As System.Double, _
    ByVal Z2 As System.Double, _
    ByVal Val5 As System.Boolean _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Val1 As System.Double
Dim Val2 As System.Double
Dim Z1 As System.Double
Dim Val3 As System.Double
Dim Val4 As System.Double
Dim Z2 As System.Double
Dim Val5 As System.Boolean

```

```
instance.SketchRectangle(Val1, Val2, Z1, Val3, Val4, Z2, Val5)
```

C#

```

void SketchRectangle(
    System.double Val1,
    System.double Val2,
    System.double Z1,
    System.double Val3,
    System.double Val4,
    System.double Z2,
    System.bool Val5
)

```

C++/CLI

```

void SketchRectangle(
    & System.double Val1,
    & System.double Val2,
    & System.double Z1,
    & System.double Val3,
    & System.double Val4,
    & System.double Z2,
    & System.bool Val5
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[Val1](#)
Upper-left x value in meters

[Val2](#)
Upper-left y value in meters

[Z1](#)
Upper-left z value in meters

[Val3](#)
Lower-right x value in meters

[Val4](#)
Lower-right y value in meters

[Z2](#)
Lower-right z value in meters

[Val5](#)
Not used

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchRectangle](#).

Example

[Create Revolve Features \(VBA\)](#)
[Connect to SOLIDWORKS Session \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchRectangleAtAnyAngle Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0